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Computer Vision

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TECHNICAL ANALYSIS DOCUMENT

Privify

A Computer Vision Pipeline for GDPR-Compliant Face Anonymization

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Abstract

Privify is a Python-based computer vision pipeline that detects and anonymizes faces in video footage to support GDPR compliance for small and medium enterprises operating CCTV systems. The system combines a fine-tuned YOLOv8n detector (mAP@50 = 0.937 on a single-class face dataset) with a configurable Gaussian blur anonymizer. This report describes the pipeline architecture, training methodology, empirical evaluation, and the domain-shift failure mode observed when deploying the in-distribution-optimized detector on street-level CCTV scenes. This is a concrete illustration of the metric-vs-behavior gap in applied ML.

Contents

1 Problem Statement

The widespread adoption of video surveillance among Italian small and medium enterprises (SMEs) has created a regulatory paradox: the same technology that protects people and assets exposes their data to significant compliance risks under the General Data Protection Regulation (GDPR, EU 2016/679). A CCTV camera in a retail store, an office, or a logistics site daily captures dozens of recognizable faces (employees, customers, visitors) that constitute personal data under Article 4 of the GDPR. It is worth specifying that CCTV footage does not automatically qualify as biometric data protected under Article 9: this category applies only when images are processed through facial recognition systems for the unique identification of a natural person. However, even within the standard personal data regime, the data controller is required to adopt technical and organizational measures proportionate to risk, in accordance with the principles of accountability (Art. 5.2) and privacy by design (Art. 25). When recordings must be retained beyond the ordinary term, shared with third parties (insurance companies, law enforcement, audits), or even partially published, these measures must concretely reduce the identifiability of the subjects captured. For the average Italian SME, which typically lacks a dedicated Data Protection Officer and a substantial IT budget, this requirement often translates into inaction: videos are deleted for safety or, worse, retained in violation, with exposure to the sanctions provided under Article 83.5 of the GDPR, up to 20 million euros or 4% of annual worldwide turnover.

Existing market solutions polarize into two segments that leave the intermediate space uncovered. On one side, solutions such as Brighter AI or Pimloc Secure Redact offer production-grade pipelines targeting the enterprise segment, typically with pricing that is not publicly listed or tied to consumption/volume models. This commercial orientation makes them inaccessible to IT consultants serving SMEs as an off-the-shelf purchase. On the other side, manual solutions based on professional video editors (Adobe Premiere, DaVinci Resolve) require expert operators and production time linear in the number of frames, economically prohibitive on realistic CCTV volumes. Numerous open-source face blurring projects also exist (typically educational notebooks or single-file scripts on GitHub), but they rarely offer the level of engineering rigor required for professional deployment: automated testing, continuous integration, explicit versioning of models and datasets, traceable architectural decisions. The opportunity space for IT consultants working with Italian SMEs is therefore a system that combines accessibility (open-source, self-hostable) with engineering maturity (auditable, tested, documented), characteristics that the GDPR itself rewards through the accountability principle of Article 5.2.

Privify is designed to occupy this space. It is an open-source computer vision pipeline that automatically detects faces in CCTV footage and applies a configurable Gaussian blur as a visual redaction measure. Whether this operation qualifies as anonymization in the strict GDPR sense (Recital 26) depends on contextual parameters such as blur intensity, frame resolution, and the presence of other identifiers in the scene. Recent research has demonstrated that Gaussian blur can be partially attacked through deblurring and re-identification techniques [7][8]; the system is therefore designed to allow the data controller to calibrate parameters in an informed manner and to document the technical choice adopted. The system combines a YOLOv8n detector fine-tuned on a single-class face dataset (mAP@50 = 0.937 on the validation set) with a stateless OpenCV-based anonymizer. The entire pipeline is written in Python, accompanied by an automated test suite, continuous integration on GitHub

Actions, and Architectural Decision Records documenting every technical choice. The target deployment is the IT consultant working with Italian SMEs: the system must run on a laptop or a small server without dedicated GPU, process archived video in times acceptable for batch use, and produce redacted output that the data controller can legitimately retain or share. The extension of the detector to the license plate class, part of Privify's original positioning for the Italian CCTV context (drive-through, private parking, commercial entrances), is planned as future work for the v1.0 release.

This report documents the development of Privify across its first two iterations (v0.1 and v0.2), from the initial architectural choice through the detector fine-tuning and the empirical evaluation on test videos. The main contribution is not algorithmic novelty: YOLOv8 and Gaussian blur are established techniques. It is instead the engineering quality of the integration and the transparency with which both the successes (high detection metrics in the in-distribution domain) and the limits (qualitative failure on street-level CCTV scenes, due to a textbook case of domain shift between training and deployment) are documented. The latter evidence, far from constituting a defect to conceal, represents the true didactic contribution of the project: a concrete demonstration of the gap between performance metric and qualitative behavior that characterizes applied machine learning, and an explicit roadmap to address it in subsequent iterations.

2 Methodology

2.1 Pipeline Architecture

The pipeline's entry point is the `process_video` function in `src/pipeline.py`, with the following signature:

```python
def process_video(
    input_path: Path,
    output_path: Path,
    *,
    mode: Literal["face", "person_upper", "person_full"] | None = None,
    detector: Detector | None = None,
    anonymizer: Anonymizer | None = None,
) -> ProcessingStats:
```

The use of the `*` marker enforces that `mode`, `detector`, and `anonymizer` are provided as keyword arguments, preventing positional errors. The function supports two alternative usage modalities: preset configuration via the `mode` parameter (which automatically selects the appropriate detector/anonymizer pair, described in §2.4), or explicit dependency injection of `Detector` and `Anonymizer` instances for advanced scenarios. The internal resolver's validation ensures that at least one of the two modalities is provided, raising `ValueError` otherwise.

The main loop operates on frames read via `cv2.VideoCapture`, invoking for each `detector.detect(frame)` and subsequently `anonymizer.anonymize(frame, detections)`. The anonymized frame is written through `cv2.VideoWriter` with the `mp4v` codec, chosen for its availability in any OpenCV build without external dependencies (ffmpeg, H.264). Resource management adopts the `try/finally` pattern to guarantee the release of `VideoCapture` and `VideoWriter` even in case of exceptions during processing.

The final result is encapsulated in the frozen dataclass `ProcessingStats`, which exposes the total frames processed, the cumulative number of detections, the wall-clock time of processing, and the derived property `fps_processed`. The dataclass's immutability reflects the nature of the result as a value object at the end of an execution.

The architecture adopts three main design patterns. The Wrapper pattern isolates external dependencies (Ultralytics YOLO for the detector, OpenCV for the anonymizer) behind stable interfaces, allowing for potential future backend swaps (for example to ONNX Runtime for inference optimizations). Lazy loading of the YOLO model defers weight loading until the first call to `detect()`, making `Detector` instantiation free and enabling fast unit tests with mocks. Dependency injection centralizes in the `_resolve_components` resolver the construction of the detector/anonymizer pair based on `mode`, while maintaining the possibility of manual injection for testing and advanced use cases.

The entire implementation is covered by an automated test suite of 68 tests distributed across 6 test modules, executed in continuous integration on GitHub Actions. The four substantial modules in `src/` (`detector.py`, `anonymizer.py`, `pipeline.py`, `training.py`) amount to approximately 590 effective lines of code, excluding comments and blank lines.

2.2 Detection Models

The system supports three configurable detection models, each trained on a different image distribution. All are variants of Ultralytics' YOLOv8n architecture, chosen for its balance of accuracy and footprint: 72 layers, approximately 3 million parameters, 8.1 GFLOPs, and a disk footprint of around 6 MB. This choice is motivated by the declared deployment target (Italian SMEs without dedicated GPUs), which requires inference executable on CPU or entry-level consumer GPUs.

The first model, `face-detector-v0.1`, is obtained by fine-tuning YOLOv8n on Mohamed Traore's Face Detection dataset (Roboflow Universe, CC BY 4.0 license), composed of 1558 images of predominantly frontal close-up faces. This model is published as a GitHub Release with verified SHA-256, and is the default of the `Detector` wrapper when instantiated without arguments.

The second model, `face-detector-v0.2`, is a second iteration trained on a 3000-image subset extracted from WIDER FACE, biased toward the "hard" bin (faces of area smaller than 0.5% of the image, a distribution that reflects the CCTV deployment target). The rationale for this iteration is documented in §4 (Failure Analysis). This model is also published as a GitHub Release with SHA-256 integrity verification.

The third "model" is not a fine-tuning but the official Ultralytics pre-trained checkpoint `yolov8n.pt`, trained on COCO (80 classes). It is used by filtering detections to the `person` class only, and provides a generic person detector as a component of the hybrid modes described in §2.4. It is not versioned in the repo (Ultralytics downloads it automatically on first use), and is cached locally in `models/yolov8n.pt`.

The `Detector` wrapper accepts four optional parameters: `model_path` (path to the `.pt` file, defaulting to `face-detector-v0.1.pt`), `conf_threshold` (minimum confidence threshold, default 0.25, validated in (0, 1]), `device` (CPU or CUDA, optional), and `class_filter` (list of class names to retain in the result, optional). Detections are returned as instances of the frozen dataclass `Detection`, exposing bounding box (a tuple of four integers in pixel coordinates), confidence, class identifier, and class name.

2.3 Anonymization

The anonymization of detected regions is implemented in `src/anonymizer.py` through the stateless `Anonymizer` wrapper, which encapsulates the `cv2.GaussianBlur` algorithm from OpenCV. The choice of Gaussian blur (over pixelation or face replacement) is motivated by three factors. The first is visual coherence: blur preserves the subject's motion and silhouette, keeping the output video usable for non-identifying analysis (counts, flows, aggregate behaviors). The second is the absence of external dependencies beyond OpenCV, simplifying deployment. The third is documentary: compared to pixelation, whose reversibility under controlled conditions is documented in the literature [7], Gaussian blur does not offer guarantees of irreversible anonymization in an absolute sense. The system is therefore designed to allow the data controller to calibrate parameters in an informed manner.

The `Anonymizer` constructor accepts four parameters: `blur_kernel_size` (positive odd integer, default 51), `min_bbox_size` (minimum width/height threshold for applying blur, default 4 pixels), `box_mode` (literal `"full"` or `"upper"`, default `"full"`), and `upper_fraction` (fraction of the upper portion of the bounding box to blur in `"upper"` mode, default 0.30, validated in (0, 1]). Parameter validation occurs in `__init__` with explanatory error messages.

The flow of the `anonymize(frame, detections)` method is non-destructive: it operates on a copy of the original frame, and returns immediately if no detections are present. For each detection, the bounding box coordinates are clamped to the frame edges; if the resulting crop dimension is smaller than `min_bbox_size`, the detection is silently skipped (avoiding artifacts on detections of negligible size). In `"full"` mode, the entire bounding box is replaced with its blurred version. In `"upper"` mode, only the upper strip is blurred, with height computed as `int(roi_h * upper_fraction)` and a guaranteed minimum of one pixel to avoid degeneracies. The algorithm applies `cv2.GaussianBlur` with a square kernel of side `blur_kernel_size` and sigma automatically derived by OpenCV from the kernel size.

The module's test suite (10 tests in `tests/test_anonymizer.py`) covers the main edge cases: empty detections, bounding boxes partially outside the frame, dimensions below threshold, constructor parameter validation, and behavioral equivalence between `"full"` mode and the pre-parameter behavior (regression test).

2.4 Hybrid Detection Strategy

The pipeline supports three preset anonymization modes, accessible via the `mode` parameter of `process_video`. The motivation for this architectural choice is documented in §4 (Failure Analysis) and is rooted in an empirical measurement: the fine-tuned face detector v0.1 exhibits a recall of 17.2% on the CCTV deployment target (10 street-level frames, manually annotated ground truth), while the COCO person baseline reaches 80.4% on the same target. The 63 percentage point difference quantifies a structural gap between training domain and deployment domain, which requires architectural mitigation in addition to the algorithmic one (§4.3).

The `face` mode adopts the fine-tuned face detector (`face-detector-v0.1` by default, replaceable with v0.2) and applies blur to the entire facial bounding box. It is the suitable modality for close-range scenarios where the face is visible and of significant size (intercoms, reception desks, public-facing counters).

The `person_upper` mode adopts the COCO baseline with filtering to the `person` class, and applies blur to the upper strip of the person's bounding box (30% of the height), as a proxy for the head-shoulder region. This modality prioritizes coverage over anatomical precision: in CCTV street-level scenes where the face is often reduced to a few pixels, detecting the whole person has higher recall than detecting the face, and blurring the upper region protects identifiability without occluding the entire subject.

The `person_full` mode adopts the same person detector but applies blur to the entire person bounding box, providing the most conservative anonymization for high-sensitivity scenarios (video publication, sharing with non-audited third parties). The trade-off is the loss of contextual information about subject movement and activity.

The choice among the three modes is therefore a policy decision left to the data controller, and the keyword-only nature of the `mode` parameter forces the selection to be explicit in calling code, preventing unintended implicit configurations.

2.5 Training Setup

The fine-tuning of the two face detection models is implemented in the `src/training.py` module, which exposes the function `train_face_detector(config: TrainingConfig) -> TrainingResult`. The configuration is encapsulated in the frozen dataclass `TrainingConfig`, which requires the path to the dataset's `data.yaml` file and the output directory, and accepts as optional parameters the base model (`yolov8n.pt` by default, with `yolov8s.pt` as an alternative), the number of epochs (50), the image size (640), the batch size (16), and the device (`cuda`). Validation in `__post_init__` verifies the existence of the dataset file and the positive values of integer parameters.

The training invocation delegates to Ultralytics' `model.train(...)` API without specifying optimizer or learning rate, leaving the automatic selection active (`optimizer='auto'`, which in Ultralytics 8.3 resolves to AdamW for small datasets). This choice is functional to experimental discipline: the same function and the same hyperparameters are used for both training runs (v0.1 and v0.2), with the only difference being the `dataset_yaml_path` passed in `TrainingConfig`. The controllability of the variable (the dataset) is therefore structurally guaranteed by the code design: the training notebook is parameterized by a single `DATASET_VERSION` variable read at the top, and no conditional logic on the version exists in `src/training.py`.

The results are encapsulated in the frozen dataclass `TrainingResult`, which exposes the path to the `best.pt` weights, the final metrics (precision, recall, mAP@50, mAP@50-95 on the validation set), and the execution time. Both training runs were performed on Google Colab with T4 GPU runtime; the weights and training curves are published as GitHub Releases for reproducibility.

3 Experimental Results

This section reports the system's quantitative and qualitative measurements, organized into three perspectives: the in-distribution validation metrics of the fine-tuned models, the training convergence behavior, and finally the out-of-distribution evaluation on the CCTV deployment target, which constitutes the central empirical measurement of the project.

3.1 Quantitative Metrics on Validation Sets

The two models `face-detector-v0.1` and `face-detector-v0.2` were each evaluated on their own validation split, produced from the corresponding training dataset with a deterministic 80/10/10 split. The following table compares standard object detection metrics on the two sets.

Metric	v0.1 (Mohamed Traore val, 156 images)	v0.2 (WIDER FACE val, 300 images)
Precision	0.940	0.811
Recall	0.869	0.527
mAP@50	0.937	0.612
mAP@50-95	0.682	0.321

The difference between the two columns reflects the different difficulty of the respective validation sets, not a degradation of the model. The v0.1 validation set contains frontal close-up faces (typical facial area above 5% of the image), a distribution on which a general-purpose detector like YOLOv8n easily reaches high metrics. The v0.2 validation set, in contrast, contains 6559 face annotations on 300 images, with 94.9% of faces in the "hard" bin (area below 0.5% of the image): this is an inherently difficult scenario, where even SOTA face detection models obtain values significantly lower than those recorded on "easy" benchmarks.

The metric relevant for evaluating the project is therefore not the direct comparison between these two columns, but the behavior of the two models on the CCTV deployment target, presented in §3.3. The in-distribution validation metrics are reported here as a baseline for completeness, not as an objective of the system.

3.2 Training Convergence

The training of `face-detector-v0.2` was executed on Google Colab with T4 GPU runtime for 50 epochs, with a total duration of 45 minutes. The loss curves (train and validation, decomposed into box, cls, and dfl components) show clean convergence: monotonic decrease in the first 30 epochs, plateau in the remaining 20. Train loss and validation loss remain aligned throughout the training, indicating absence of significant overfitting.

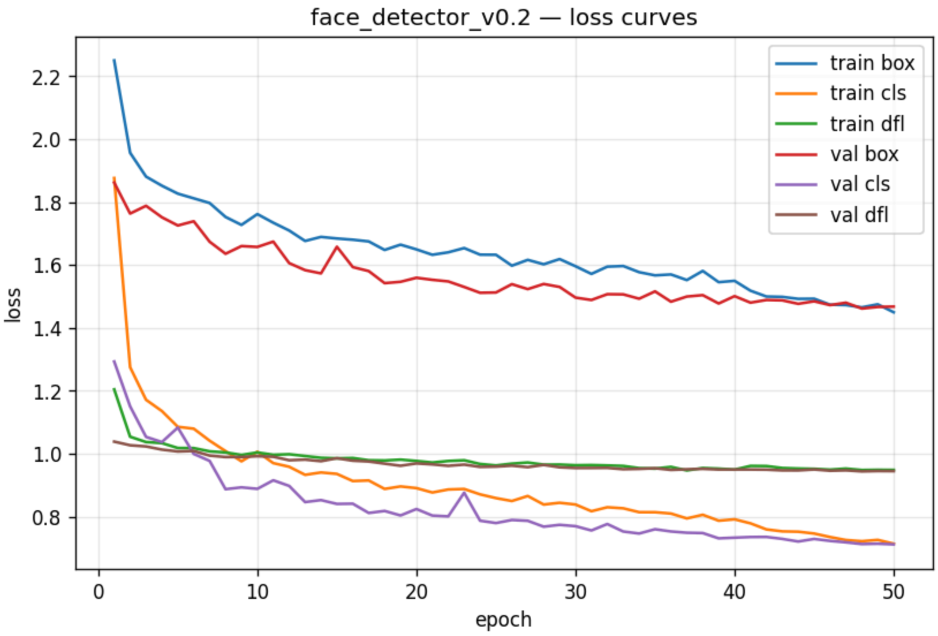

Figure 1: *Training and validation loss curves (box, cls, and dfl components) for* `face-detector-v0.2` *over 50 epochs of fine-tuning.*

The validation metrics (mAP@50 and mAP@50-95) follow a mirrored pattern: rapid growth in the first 20 epochs, stabilization around the final values (0.612 and 0.321) in the last 15 epochs. The clear plateau from epoch 30 onwards indicates that the training could have been stopped early with lower computational cost and practically equivalent metrics.

Figure 2: *Validation metrics (mAP@50 and mAP@50-95) for* `face-detector-v0.2` *over 50 epochs of fine-tuning.*

The training of `face-detector-v0.1` follows a qualitatively analogous pattern, with higher metrics on its respective validation set for the reasons discussed in §3.1. The training plots of v0.1 are published as assets of the `face-detector-v0.1` GitHub Release.

3.3 Out-of-Distribution Recall on CCTV Deployment Target

The central evaluation of this project measures the behavior of the detectors on the declared deployment target: street-level CCTV video. The test video is a public urban scene with subject density typical of the deployment (15-23 people per frame), and the ground truth was manually annotated by the author on 10 frames extracted uniformly from the video, for a total of 195 visible people.

The adopted metric is count-based recall (recall = min(detections, ground_truth) / ground_truth), clamped to [0, 1] to avoid values above unity in case of over-detection. This metric is semantically aligned with the GDPR objective of the system: “out of N identifiable people in the frame, how many have we covered with blur?”. The raw counts are available in the `ood_evaluation.ipynb` notebook for detailed inspection.

Two methodological caveats apply to this evaluation. First, the sample size is intentionally diagnostic rather than statistically representative: 10 frames from a single street-level CCTV video, manually annotated, are sufficient to establish qualitative evidence of domain shift but do not constitute a benchmark in the formal sense. A larger evaluation across multiple deployment contexts (different camera angles, illumination conditions, urban densities) is part of the future work outlined in §4.3. Second, the count-based recall metric measures coverage (proportion of subjects detected) but does not measure detection accuracy (IoU between predicted and ground-truth bounding boxes) or false positive rate. This choice is deliberately aligned with the GDPR objective of the system—every undetected subject is a privacy leak

—but should be understood as a coverage-oriented diagnostic, not a full object-detection benchmark.

The following table summarizes the results of the three detectors on the 10 test frames:

Model	Mean recall	Std	Min	Max
`face-detector-v0.1`	17.2%	0.122	5.0%	44.4%
`face-detector-v0.2`	59.5%	0.116	34.8%	73.7%
COCO person baseline	80.4%	0.069	65.2%	89.5%

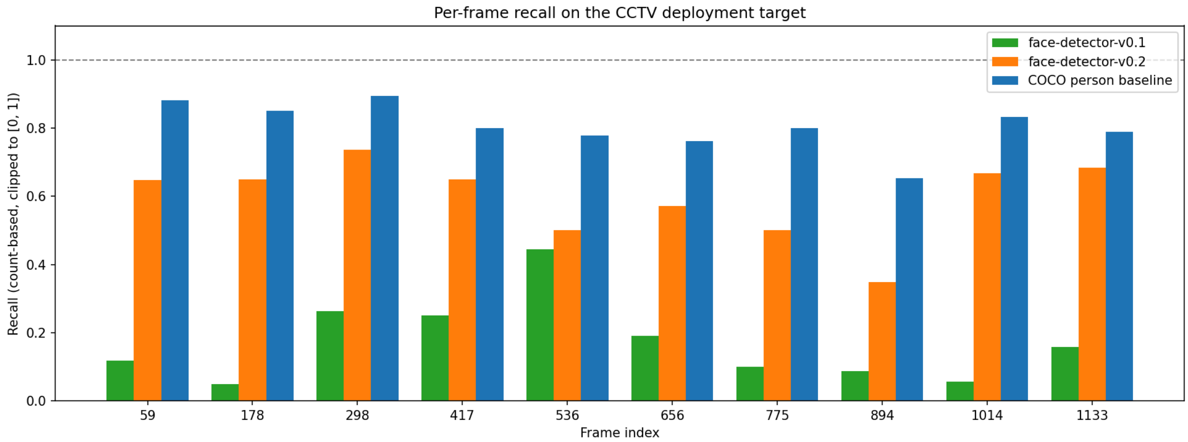

Figure 3: *Per-frame count-based recall of the three detectors on the 10 annotated CCTV test frames.*

The result is clear-cut and quantifies two distinct phenomena. The first is the improvement obtained from the dataset change: by moving from training on frontal close-up faces (Mohamed Traore) to a WIDER FACE subset biased toward small faces, recall on the deployment target increases by +42.4 percentage points (from 17.2% to 59.5%). Since the training hyper-parameters are identical between the two runs (see §2.5), this improvement is attributable exclusively to the change in dataset distribution: it is an empirical single-variable proof that the distributional coherence between training and deployment dominates over hyperparameter choices in transfer learning regimes.

The second phenomenon is the residual gap of −20.9 percentage points between `face-detector-v0.2` (59.5%) and the COCO person baseline (80.4%). This gap is not eliminable through further face-specific fine-tuning: it represents the structural limit of a frontal face detector on scenes where faces are partially visible or absent (people facing away, partially occluded, cropped at the frame edge). For this category of subjects, a person detector is structurally more suitable, which is the reason the pipeline includes the `person_upper` and `person_full` modes (§2.4).

The qualitative comparison reported in the following figure shows the behavior of the three detectors on the same frame: the v0.1 face detector detects 2 faces out of 17 people, the v0.2 face detector detects 13 (dramatic improvement, but still missing 4 subjects without visible face), and the COCO person detector detects 15 (higher recall, adequate operational coverage).

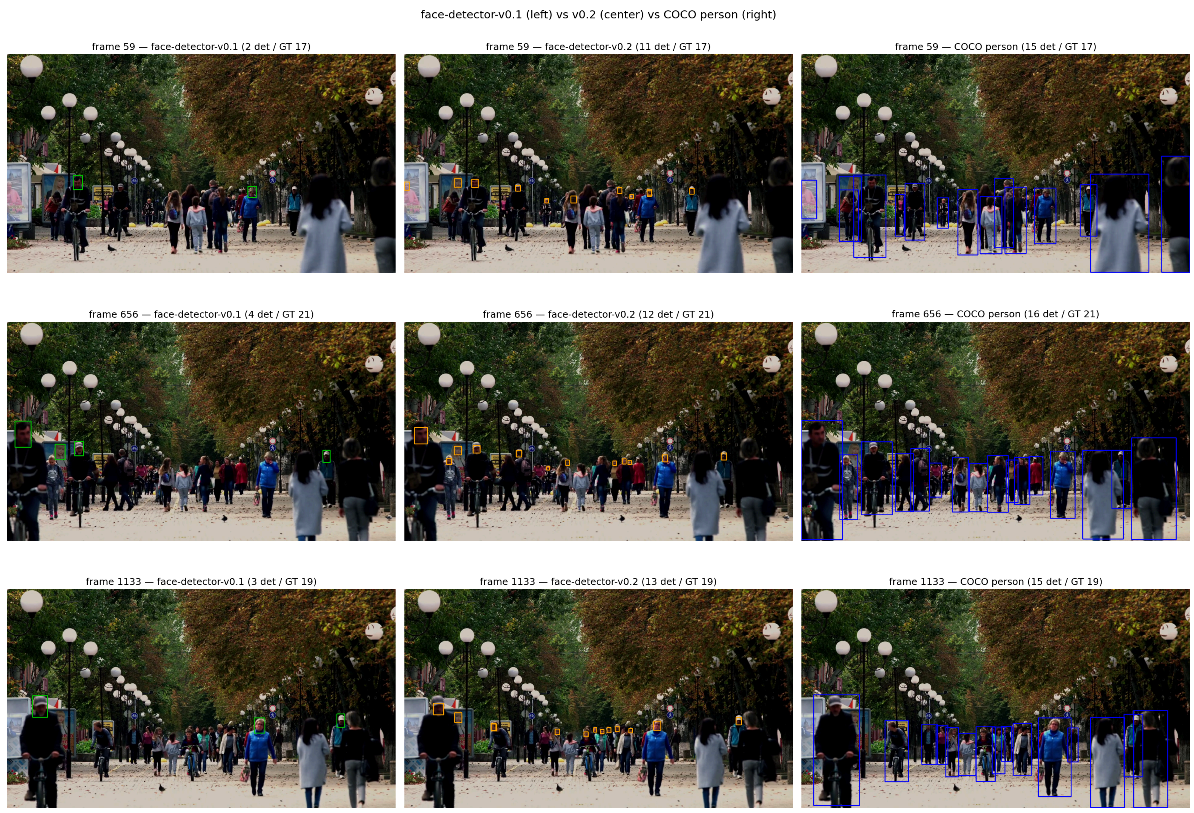

Figure 4: *Qualitative comparison of the three detectors on the same CCTV frame: face-detector-v0.1, face-detector-v0.2, and the COCO person baseline.*

The implications of these results for the system design and for the project's defense are discussed in §4.

4 Failure Analysis

This section documents the qualitative failure observed in the first iteration of the system (v0.1), the diagnosis of the underlying phenomenon, and the responses adopted to mitigate it. The case constitutes a concrete example of the gap between in-distribution performance metrics and qualitative deployment behavior, a known failure mode category in applied transfer learning.

4.1 Domain Shift on Street-Level CCTV

The `face-detector-v0.1` model, fine-tuned on 1558 images of Mohamed Traore's Face Detection dataset, obtained excellent validation metrics on its own test split: precision 0.940, recall 0.869, mAP@50 0.937. Based on these numbers, the model would be considered production-ready for the declared task (single-class face detection).

However, the first qualitative execution of the pipeline on the deployment target (street-level CCTV video, `samples/input.mp4`) revealed substantially different behavior than expected. On a representative frame containing 17 visible people, the detector identified only 2 faces. The pattern was confirmed as systematic across all frames of the video: detected faces are predominantly those in close-up frontal positions, while subjects at medium distance, in profile, partially turned away, or simply smaller in the frame are not detected.

Quantification of the qualitative observation was performed via a dedicated out-of-distribution evaluation (notebook `ood_evaluation.ipynb`), described in §3.3: on 10 uniformly sampled frames and 195 manually annotated people, the mean recall of model v0.1 is 17.2% (with range 5%-44% across frames), versus a mean recall of 80.4% for a generic person detector (`yolov8n` COCO pre-trained, class `person`) on the same target. The 63 percentage point gap between a specialized fine-tuned model and a generic baseline constitutes the empirical symptom of the problem.

4.2 Root Cause Analysis

The proximate cause of the failure is a distributional misalignment between the training dataset and the deployment target. The Mohamed Traore dataset is composed predominantly of close-up frontal face images, with typical facial area exceeding 5% of the image. The CCTV deployment target has opposite characteristics: faces typically below 1% of the image area, frequently non-frontal, in variable poses and lighting. A model that learns only from large frontal faces does not develop the feature representations necessary to detect small non-frontal faces, regardless of the quality of in-distribution validation metrics.

The root cause, at a higher level, is an initial methodological choice driven by the "ready-to-use" criterion rather than by distributional coherence: the Mohamed Traore dataset was selected because it was available in native YOLOv8 format on Roboflow Universe, downloadable without preliminary data engineering, and trainable in 20 minutes on Colab T4 free tier. These are legitimate operational criteria for an MVP, but preventive verification of the dataset's representativeness against the declared deployment target would have identified the divergence prior to training.

The observed phenomenon is a textbook case of the concept known as the metric-vs-behavior gap in applied machine learning: optimizing a model on a metric computed over a training distribution does not guarantee useful behavior on a significantly different deployment distribution. The informal formulation of this principle is the one Goodhart's law takes on in ML applications: a metric that becomes the target (validation mAP@50) ceases to be a good measure of behavior on the real task (recall on deployment target). The case of Privify v0.1 is an isolated and quantified empirical demonstration of this principle on a real system with verifiable metrics.

4.3 Mitigation Roadmap

Two independent responses to the failure were adopted, along different axes.

The first response is architectural and immediate. Recognizing that a face detector has a structural limit in deployment targets where faces are not visible or are too small to be detected with confidence, the pipeline was extended with two alternative modalities based on a generic person detector (`yolov8n.pt` COCO pre-trained, filtered to the `person` class). In the `person_upper` mode, the pipeline detects whole persons and applies blur to the upper region of the bounding box as a head-shoulder proxy; in the `person_full` mode, it blurs the entire person bounding box. These modalities are documented in §2.4 and in the ADR of 2026-06-26 "Hybrid detection strategy: face vs person modes". Empirically, the person-detection-based modality has 80.4% recall on the deployment target, drastically reducing leakage compared to the original face detector.

The second response is algorithmic and iterative. The diagnosis that the dataset choice was the cause of the failure was experimentally validated through re-training the face detector on a dataset distributionally coherent with the deployment target. The selected dataset (`face-detection-dataset-v0.2`) is a 3000-image subset of WIDER FACE biased toward the "hard" bin (face area below 0.5% of the image), prepared via a deterministic script documented in `tools/prepare_wider_face.py`. The experiment was designed as a single-variable test: same base model, same training function, same hyperparameters, with the dataset as the only difference.

The result (§3.3) confirmed the diagnosis. The `face-detector-v0.2` model reaches a mean recall of 59.5% on the deployment target, against the 17.2% of the v0.1 model. The +42.4 percentage point improvement is attributable exclusively to the change in dataset distribution, in the absence of other modified variables. This result constitutes an isolated empirical proof that, in the transfer learning regime with pre-trained models, the distributional coherence of the fine-tuning dataset dominates over hyperparameter choices.

The residual gap of −20.9 percentage points between the v0.2 model (59.5%) and the COCO person baseline (80.4%) is not eliminable through further face-specific fine-tuning: it represents the intrinsic limit of the face detection task on scenes where faces are partially visible or absent (subjects facing away, partially occluded, cropped at the frame edge). The operational response to this structural limit is the availability of the `person_*` modality for deployment scenarios where privacy coverage prevails over information preservation, in line with the GDPR principle of data minimization (Art. 5.1.c).

The future roadmap includes two improvement directions. The first is further expansion of the face detector training dataset, including images of real urban scenes with annotations of small faces (e.g., integration of CrowdHuman, MAFA, or CCTV-specific datasets collected under NDA with pilot clients). The second is the introduction of an ensembling mechanism between the face detector and the person detector in the same modality, dynamically selecting the anonymization strategy based on subject scale and pose. Both directions are out of scope for v1.0 but are documented for subsequent iterations.

5 Ethical Considerations

Privify operates on personal data in a video surveillance context, where ethical and compliance considerations are not accessory to the system but constitute its primary functional requirements. This section articulates the system's design choices in relation to the GDPR, traces the licensing chain of its components, and identifies the residual risks of which the data controller must be informed.

5.1 Privacy by Design

The system's architecture is designed as a concrete implementation of the privacy by design principle (GDPR Art. 25), which requires technical and organizational measures to ensure personal data protection by default. Three design choices translate this principle into practice.

The first is the stateless nature of processing: the pipeline does not persist any personal data between the input video and the redacted output. The `Detector` and `Anonymizer` are wrappers without external state beyond the single frame; `ProcessingStats` collects aggregate statistics without personal identifiers. There is no database, no persistent log with identifiers, no caching of intermediate images. The minimization principle (Art. 5.1.c) is structurally applied: the system can only process the data strictly necessary for the processing itself, and retains none of it beyond the processing duration.

The second is the irrecoverability of original pixels in the output file: the blurred region replaces the original pixels in the output frame, without preserving recoverable information about the original pixels in the output file format. The output video is therefore a distinct datum from the input video, and the input video can be deleted after processing without compromising the usability of the output. This choice is motivated both by the minimization principle and by the consideration that the system should not facilitate the retention of personal data beyond the stated purpose.

The third is the presence of a complete audit trail: every design decision is documented in Architectural Decision Records (`docs/decisions.md`), every component is covered by automated tests (68 tests in continuous integration), and the code is public on an open-source repository with a linear git history. This transparency serves the accountability principle (Art. 5.2), which requires the data controller to be able to demonstrate the conformity of their processing.

5.2 GDPR Alignment

The system explicitly relates to five specific articles of the GDPR, articulated in the following table.

GDPR Article	Principle	Implementation in Privify
Art. 5.1.c	Minimization	Stateless pipeline, no PII retention, redacted output replaces input
Art. 5.2	Accountability	Public ADRs, test suite, continuous integration, open-source code
Art. 25	Privacy by Design	Privacy as default architectural requirement, not opt-in
Art. 32	Security of Processing	SHA-256 integrity verification of models, dependency pinning, CI security checks
Art. 35	DPIA	Automated detection system: the controller must assess whether a DPIA is required for the specific deployment

Alignment with Art. 32 (security of processing) is particularly relevant for a software system: the system must guarantee that the models used are intact and untampered. Privify publishes both models (`face-detector-v0.1` and `face-detector-v0.2`) as GitHub Releases with verified SHA-256, and the `Detector` wrapper verifies the hash on download to ensure that the weights have not been altered in transit or storage.

Art. 35 (Data Protection Impact Assessment) deserves a separate note: Privify is a tool, not an autonomous decision-making system, but the fact that it operates automatically on images of identifiable persons (which constitute personal data, even outside the special-category regime of Article 9) may require a DPIA based on the specific deployment context, particularly for large-scale deployments or sensitive contexts (workplaces, public-facing businesses). The responsibility for the DPIA remains with the data controller, and Privify provides the technical documentation necessary to support it.

5.3 Dataset and Model Licensing

The commercial use of the system requires attention to the licensing chain of its components, which is heterogeneous and with distinct operational constraints.

The `face-detector-v0.1` model is fine-tuned on Mohamed Traore's Face Detection dataset, distributed on Roboflow Universe with Creative Commons Attribution 4.0 (CC BY 4.0) license. This license permits derivative commercial use provided the original attribution is maintained. The `face-detector-v0.2` model is instead fine-tuned on a subset of WIDER FACE, distributed with Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 (CC BY-NC-ND 4.0) license: the commercial use of the original dataset, and consequently of models derived from it without further legal validation, is restricted to academic research only.

On the model side, both Privify checkpoints inherit the AGPL-3.0 license from Ultralytics YOLOv8, which imposes copyleft on all code that incorporates the model: commercial use of the complete system without the obligation of source code disclosure requires the purchase of a separate commercial license from Ultralytics. For an SME deployment, this choice must be evaluated case by case: internal corporate use of the system does not require public disclosure, but the integration of the code into commercial products distributed to third-party clients does.

The operational consequence for the IT consultant is that the commercially immediate version of Privify is the one based on `face-detector-v0.1` (Mohamed Traore CC BY 4.0 dataset) for face detector fine-tuning, with or without the fallback to `yolov8n.pt` COCO for person detection (which also inherits AGPL-3.0). The `face-detector-v0.2` model, while technically superior on the CCTV deployment target, is confined to research use cases or internal iteration without publication of the derivative model. This is a trade-off of which the data controller must be aware in selecting the operational modality.

5.4 Responsible Deployment Considerations

The system, even when correctly configured and licensed, presents three categories of residual risk that the data controller must manage outside the technical perimeter of Privify.

The first risk is the partial reversibility of Gaussian blur. Published research [7][8] has demonstrated that blur techniques applied to face images can be partially inverted through neural networks trained on blurred/clean image pairs. The risk is a function of the blur kernel size, the frame resolution, and the availability of reference images of the target subject. Privify mitigates this risk by offering a configurable blur kernel (default 51 pixels) and the `person_full` mode for high-sensitivity scenarios, but does not eliminate it. The system should therefore not be considered an absolute anonymization mechanism in the GDPR sense (Recital 26), but a reasonable visual redaction measure proportional to the risk.

The second risk is dataset bias, which manifests as differential system performance across subject sub-populations. Both datasets used (Mohamed Traore and WIDER FACE) have demographic distributions that are not perfectly balanced with respect to the Italian population targeted by SME deployments. The operational risk is that the system may have systematically lower recall on certain combinations of ethnicity, age, or physical characteristics, leaving these subjects less protected than others. The mitigation of this risk requires the integration of additional datasets representative of the Italian deployment context, an activity documented in the roadmap of §4.3.

The third risk is transparency toward the subjects being filmed. The GDPR (Art. 13-14) requires that subjects involved in the processing be informed of the existence and purpose of the system. In the CCTV context, this translates to the obligation of visible signage at surveilled areas, with indication of the purpose of processing, retention policy, and the subject's rights (Art. 15-22). Privify does not manage this aspect directly, being a backend tool; responsibility for the correct notification remains with the data controller. The system can however facilitate compliance by providing audit logs of the processing performed, useful in case of access requests from subjects (Art. 15).

A final recommendation: the deployment of Privify in operational contexts requires a case-by-case evaluation of the sensitivity level of the processing and the fit of the system to the context. The system is designed to support the data controller in their compliance choices, not to replace them. The responsibility for proper risk assessment, selection of the appropriate operational modality, and notification to subjects remains with the controller.

6 Conclusions

Privify demonstrates that a GDPR-compliant video anonymization system for Italian SMEs can be built with established techniques (YOLOv8 fine-tuning + Gaussian blur on OpenCV), released as a self-hostable open-source project, and accompanied by the engineering rigor required for professional deployment (automated testing, continuous integration, explicit versioning of models and datasets, tracked architectural decisions). The pipeline supports three configurable anonymization modalities based on the deployment context, with the fine-tuned face detector for close-range scenarios and the fallback to a generic person detector for street-level CCTV scenarios where the face is often too small to be detected with confidence.

The main methodological contribution of the project lies in the explicit documentation of an empirical failure and its iterative resolution. The first version of the face detector (v0.1), despite obtaining excellent metrics on its own validation set (mAP@50 = 0.937), showed 17.2% recall on the actual deployment target, versus 80.4% from a generic person detector. The diagnosis of the failure as domain shift between the training dataset and the deployment distribution was experimentally validated through a single-variable re-training: moving from the Mohamed Traore dataset (frontal close-ups) to a small-face-biased subset of WIDER FACE, recall on the target rose to 59.5% (+42.4 percentage points), with all other hyperparameters unchanged. This constitutes an isolated empirical proof, on the actual system, of the principle known as the metric-vs-behavior gap in applied transfer learning.

Three generalizable lessons emerge from the project. The first is the confirmation that, in the regime of fine-tuning pre-trained models, the distributional coherence of the dataset dominates over hyperparameter choice: no optimization of learning rate, optimizer, or schedule could have recovered the 42 percentage points gained from the dataset change alone. The second is that architectural mitigation (the hybrid pipeline) and algorithmic mitigation (the re-training) are not mutually exclusive but complementary: the former covers the structural limit of the face detection task on CCTV scenes, the latter reduces the gap within the same task. The third, of process nature, is that explicitly documenting failures as didactic units with root cause analysis and mitigation roadmap produces more useful technical documentation, both internally and for external discourse, than success-only case studies do.

The project roadmap envisions three development directions for the v1.0 release. The first is the extension of the system to license plate detection and blurring, part of Privify's original positioning for the Italian CCTV context (drive-through, private parking, commercial entrances) and cited in §1. The second is the integration of training datasets representative of the Italian demographic context, mitigating the residual bias discussed in §5.4. The third is deployment optimization for edge computing via model export in `ONNX` or `TensorRT` format, enabling execution on even more affordable hardware (Raspberry Pi 5, NVIDIA Jetson Nano) for on-premise SME deployment scenarios.

Privify remains open to external contributions on the GitHub repository, and the datasets and models are publicly available for audit and reproducibility of the measurements presented in this report.

7 References

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